

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.2: Animal Transport — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 3.2: Animal Transport
Driving Question	DRIVING QUESTION: How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon: Why Does Kipchoge Recover So Fast?	Learners observed two heart-rate recovery curves — one from elite marathon runner Eliud Kipchoge and one from a typical teenager after jogging 400 m — and heard a comparative introduction to the transport systems of a grasshopper, a tilapia from Lake Victoria, a frog from the Rift Valley wetlands, and a mammal.	Learners articulated initial noticings and predictions about why transport systems differ across animals, and began to connect the idea that different animals have different internal structures for moving substances around their bodies.	Learners produced Iteration 1 of their personal transport system model and generated genuine wonder-questions for the Driving Question Board, establishing the inquiry questions that will be investigated across all twelve lessons of the sub-strand.
Lesson 2 Why Can't Large Animals Just Use Diffusion? — The Importance of Transport Systems	Learners measured and recorded their own pulse rate (BPM) and breathing rate at rest, immediately after 2 minutes of star jumps, and after 5 minutes of recovery. They also calculated the surface-area-to-volume (SA:V) ratio for cubes of three different sizes (1 cm, 2 cm, 3 cm sides) and compared their pulse-rate recovery curve to Eliud Kipchoge's published post-race heart-rate data displayed on the board.	As the size of an organism increases, its surface-area-to-volume ratio decreases — meaning a smaller proportion of its body surface is available for diffusion relative to its volume. Diffusion alone is too slow and too limited in range to supply oxygen and nutrients to — or remove waste from — the deep cells of a large, complex animal. Large animals therefore require a dedicated transport system to: (1) deliver oxygen and nutrients to all cells; (2) remove metabolic wastes (CO ₂ , urea); (3) transport hormones; (4) regulate body temperature; and (5) circulate immune cells. These needs are dramatically amplified during exercise, as Kipchoge's and learners' own pulse-rate data demonstrate.	Why Kipchoge cannot survive on diffusion alone: his body is far too large — muscle cells lie centimetres away from any surface, and diffusion over such distances would take hours, far too slow to sustain marathon pace. The data from the SA:V investigation showed that as organisms grow larger, the ratio of surface area to volume falls sharply. A single-celled amoeba (tiny, high SA:V) can exchange all substances directly through its cell membrane; Kipchoge's quadriceps muscles (deep, low SA:V from the skin) require a pumped circulatory system to bridge that distance in seconds. This is why EVERY large, complex animal — fish, amphibian, reptile, bird, mammal — has evolved some form of transport system,

			even though the structures differ enormously.
Lesson 3 Transport in Insects: The Open Circulatory System of the Grasshopper	A grasshopper was examined (dissection or diagram). Learners observed: a dorsal vessel running along the grasshopper's back with small openings called ostia; an open body cavity (haemocoel) with no blood vessels branching to individual organs; haemolymph that is pale yellow-green in colour, NOT red.	Insects have an OPEN circulatory system: haemolymph is pumped by the dorsal vessel through ostia and floods the haemocoel, bathing organs directly. Unlike human blood, haemolymph does NOT carry oxygen — insects use a separate tracheal system for gas exchange. This means an insect's transport system operates at low pressure and cannot direct oxygen-rich fluid to specific muscles on demand.	The grasshopper survives without a powerful heart because its small body size means diffusion distances are short, and its tracheal system delivers oxygen directly to cells without using the circulatory system. Eliud Kipchoge could NOT survive on a grasshopper's circulatory system because his large body and extreme muscular demands during a marathon require a high-pressure, closed system that can rapidly direct oxygenated blood to working muscles — something an open haemocoel system cannot do. Today's evidence updates our driving question: the STRUCTURE of the transport system (open vs. closed, presence or absence of oxygen-carrying fluid) directly determines the level of physical performance an animal can sustain.
Lesson 4 Transport in Fish — The Single Circulatory System of the Tilapia	Learners traced blood flow through a colour-coded diagram of the tilapia circulatory system, identifying the two-chambered heart, the location of gill oxygenation, the path of the dorsal aorta to body tissues, and the pressure drop that occurs after blood passes through the gill capillaries.	Learners learned that the tilapia has a CLOSED but SINGLE circulatory system with a two-chambered heart (one atrium, one ventricle). Blood travels in one loop: heart → gills (oxygenated) → body tissues (oxygen delivered) → back to heart. Because blood passes through TWO sets of capillaries (gills then body) before returning to the heart, pressure drops significantly — making delivery to body tissues less forceful than in a mammal's double circulatory system. This makes the fish system more efficient than an insect's open system, but less efficient than Kipchoge's four-chambered double circulatory system.	The structural difference between the fish's single circuit and Kipchoge's double circuit explains why the tilapia cannot sustain the high-speed, high-oxygen-demand activity that Kipchoge's cardiovascular system supports. In the double circuit, the left ventricle re-pressurises oxygenated blood before sending it to the body muscles — ensuring high-pressure, high-volume oxygen delivery. In the tilapia, no such re-pressurisation occurs. This is a structural limitation that connects directly to the driving question: transport system structure determines the animal's capacity to survive and thrive at different energy demands.
Lesson 5 Transport in Amphibians and Reptiles — The Trade-Off Heart	Learners observed colour-coded diagrams of the frog's three-chambered heart (two atria, one ventricle) and the reptilian heart with a partially divided ventricle. They traced the path of a red blood cell through the frog's circulatory system using coloured pencils, noting where oxygenated and deoxygenated blood mix in the single	Learners learned that the frog has a three-chambered heart in which oxygenated blood (from lungs and skin) and deoxygenated blood (from body tissues) mix in the single ventricle, reducing the efficiency of oxygen delivery to tissues. Reptiles have a partial septum in the ventricle that reduces — but does not	Learners explained that the mixing of oxygenated and deoxygenated blood in the frog's single ventricle means body tissues receive blood that is only partially oxygenated, limiting the animal's sustained aerobic capacity. This is why a Rift Valley frog cannot sustain high-intensity activity like marathon running. They connected this

	ventricle. They also analysed a comparative data table showing estimated % oxygenation of blood reaching body tissues across five animal groups and plotted a bar graph to visualise the efficiency differences.	eliminate — this mixing. This structural difference is a trade-off: frogs and reptiles are ectotherms with lower energy demands, so a fully separated circulation is not necessary for their survival. The four-chambered heart (as in Kipchoge) is needed only when oxygen demand is very high and sustained.	to the driving question by reasoning that Kipchoge's heart must have complete separation of the two blood types — a feature that allows near-100% oxygenated blood to reach his muscles at world-record pace. Learners updated their cumulative Animal Transport System Model to include the amphibian and reptilian tiers and revised their Efficiency Meter to show the relative oxygen delivery performance of each animal group studied so far.
Lesson 6 Transport in Mammals — The Four-Chambered Heart (Heart Dissection and Circuit Mapping)	Learners observed a real sheep or goat heart (or detailed diagram): four distinct chambers separated by a thick muscular septum; the left ventricular wall noticeably thicker than the right ventricular wall; valves between the atria and ventricles preventing backflow; the aorta and pulmonary artery leaving from the top; the vena cava and pulmonary veins entering at the top; coronary vessels on the outer surface supplying the heart muscle itself.	The four-chambered double circulation of the mammalian heart explains the efficiency advantage over the three-chambered frog heart and the two-chambered fish heart. Full separation of oxygenated and deoxygenated blood means muscles receive blood with the maximum possible oxygen concentration at every heartbeat. This is the structural basis for the high metabolic activity of endothermic mammals. The question of HOW the heart beats rhythmically and how Kipchoge controls his heart rate during and after a race remains open — this is the focus of Lesson 7.	The jump from the reptile partial septum (Lesson 5) to the complete mammalian septum (Lesson 6) explains why mammals can sustain oxygen demands far beyond what a frog or reptile could support. The driving question advances: we now have the structural evidence to explain why Kipchoge does not run out of oxygen at world-record pace. What remains is to explain the pumping mechanism, the role of the sinoatrial node in controlling heart rate, and why his recovery is so rapid — which Lesson 7 will address directly.
Lesson 7 Pumping Mechanism of the Mammalian Heart: The Cardiac Cycle, Heart Sounds, and the SAN	Learners measured their own resting and post-exercise pulse rates, plotted a heart-rate recovery curve, and compared it to Kipchoge's recovery data. They also observed the lub-dub rhythm through a desk-tapping simulation and annotated a four-panel cardiac cycle diagram showing atrial systole, ventricular systole, and diastole.	The heart pumps through a repeating three-phase cardiac cycle (atrial systole → ventricular systole → diastole). Heart sounds ('lub-dub') are caused by the closing of valves — bicuspid/tricuspid valves at the start of ventricular systole, and semi-lunar valves at its end. The sino-atrial node (SAN) in the right atrial wall generates the electrical impulse that initiates each cycle, acting as the heart's intrinsic pacemaker. Cardiac output equals heart rate multiplied by stroke volume; Kipchoge's trained heart achieves normal cardiac output at only 37 BPM because each beat pumps approximately 135 mL — nearly double that of an untrained heart.	Kipchoge's heart recovers so rapidly from 170 BPM to 37 BPM because his trained ventricles have a much larger stroke volume (~135 mL per beat vs. ~67 mL for an average person). His SAN only needs to fire 37 times per minute at rest to maintain adequate cardiac output, so once exercise demand drops, his heart rate falls quickly back to this efficient baseline. This is why a structurally superior mammalian heart — with its four chambers, powerful ventricular walls, and intrinsic SAN pacemaker — enables elite athletic performance in a way that the open circulatory system of a grasshopper, with no true heart chambers and no SAN, never could.
Lesson 8 The Lymphatic System: The Hidden Highway of Animal Transport	A demonstration showing coloured water leaking from a tube into a sponge (representing plasma leaking from capillaries into tissue spaces), and an	The lymphatic system is a one-way drainage network that collects excess tissue fluid (lymph) from intercellular spaces and returns it to the bloodstream via the thoracic	During the Berlin Marathon, the enormous blood pressure in Kipchoge's capillaries forces large volumes of plasma into his muscle tissues. The lymphatic system

	annotated diagram tracing the lymphatic pathway from lymph capillaries through lymph nodes and ducts back to the subclavian veins near the heart.	duct and right lymphatic duct. Lymph nodes filter pathogens from lymph. When lymph vessels are blocked (e.g., by parasitic worms in filariasis), fluid accumulates in tissues causing oedema. Simpler animals with open circulatory systems, like grasshoppers, do not require a lymphatic system because their haemolymph bathes tissues directly without a separate capillary-tissue boundary.	drains this fluid rapidly back into the bloodstream, preventing oedema and keeping his muscles functioning at peak capacity. Without a functioning lymphatic system — as seen in patients with lymphatic filariasis in coastal Kenya — fluid accumulates in the legs causing chronic swelling. A grasshopper with an open circulatory system has no such fluid-balance problem because there is no separate plasma compartment: haemolymph simply fills all body spaces freely.
Lesson 9 The Immune System and Blood Clotting: The Body's Defence and Repair Network	Learners built a physical model of the blood clotting cascade using yarn, beads, and cotton wool; analysed a data table comparing blood counts of a healthy child versus Achieng, a 12-year-old malaria patient from Mbita on Lake Victoria; and completed a non-specific vs. specific immunity comparison table using data cards set in Kitale.	Blood performs three integrated functions within the transport system: delivery (RBCs transporting O₂ and CO₂), defence (phagocytes and lymphocytes providing non-specific and specific immunity), and repair (platelets and the fibrinogen-to-fibrin clotting cascade). Malaria caused by <i>Plasmodium falciparum</i> destroys red blood cells, lowers haemoglobin levels, and forces the heart to compensate by increasing resting heart rate — directly impairing the transport system's efficiency.	Eliud Kipchoge's extraordinary athletic performance depends not only on a powerful heart but on fully functional blood — rich in healthy red blood cells, equipped with an effective immune defence, and capable of rapid clotting at injury sites. A grasshopper's haemolymph, by contrast, lacks fibrin-based clotting and has only limited cellular immunity — sufficient for a low-metabolic-demand insect but entirely inadequate for sustaining world-record marathon performance. The mammalian transport system's complexity is inseparable from the complexity of the demands placed on it.
Lesson 10 ABO and Rhesus Factor Blood Grouping	Learners added coloured Anti-A and Anti-B indicator solutions to four simulated blood samples and observed colour changes (or no change) to determine the ABO blood group of each sample. They also read a bar chart of Kenyan blood group distribution data from the Kenya National Blood Transfusion Service and annotated a short passage on the Rhesus factor.	Learners learned that human red blood cells carry surface antigens (A, B, both, or neither) that define ABO blood group, and that plasma contains corresponding antibodies that cause agglutination when mismatched blood is introduced. They learned the rules of transfusion compatibility, that blood group O is the most common in Kenya (~46%) making it the most in-demand at the KNBTS, and that ~94% of Kenyans are Rh-positive. They also learned that insects such as grasshoppers have haemolymph rather than true blood and therefore have no ABO system.	Learners explained that the molecular identity of red blood cells is a fixed feature of the mammalian transport system that must be precisely matched in medical interventions such as transfusions. They connected this to the driving question by reasoning that the efficiency of Kipchoge's transport system depends not only on heart structure and vessel design but also on the biochemical precision of the blood itself — a level of complexity entirely absent in the open circulatory systems of grasshoppers and other invertebrates studied earlier in the sequence.
Lesson 11 Diversity of Transport	Learners observed and physically manipulated Evidence Cards containing	Learners learned that circulatory complexity increases progressively from insects	Learners explained why Kipchoge's heart sustains world-record speeds — the double

Systems: Comparative Overview	data from all previous lessons, noticing that each animal group has a distinct combination of circulation type, heart structure, transport fluid, and blood separation — and that disputed cards revealed remaining gaps in understanding.	through fish and amphibians to mammals, characterised by three evolutionary trends: open to closed systems, single to double circulation, and incomplete to complete separation of oxygenated and deoxygenated blood; they also confirmed that the mammalian four-chambered heart with full double circulation is what enables elite metabolic performance such as that of Eliud Kipchoge.	circulation delivers fully oxygenated blood to muscles at maximum efficiency without dilution — and why a grasshoppers dorsal vessel cannot do the same, because haemolymph bathes organs directly and does not carry oxygen, making it sufficient only for the grasshoppers low metabolic demands.
Lesson 12 Final Model Presentations & Celebratory Sensemaking: Answering the Driving Question	Learners observed their own Lesson 1 model drawings (redistributed by teacher) alongside their final cumulative models during the gallery walk, noting specific structural and conceptual changes. They also observed peer group final models at three gallery-walk stations, reading diagrams comparing open and closed circulatory systems across the grasshopper, tilapia, frog, and human (Kipchoge). The class observed the live-built Class Consensus Model grow element-by-element as individual learners contributed one evidence-based label or arrow each.	Learners synthesised and consolidated the following key understandings from across all 12 lessons: (1) Transport systems vary in structure — open systems (grasshopper) use haemolymph that bathes organs directly and require no closed vessels; closed systems (fish, frog, mammal) use blood confined to vessels, enabling higher pressure, faster delivery, and finer control. (2) Complexity of transport system correlates with metabolic demand — more active, larger, or more complex animals require more efficient delivery of oxygen and nutrients and removal of waste products. (3) Double circulation (mammals) separates pulmonary and systemic circuits, ensuring fully oxygenated blood reaches the body's tissues — the structural basis of Kipchoge's ability to sustain high-intensity effort and recover rapidly. (4) The four-chambered mammalian heart is a structural adaptation that prevents mixing of oxygenated and deoxygenated blood, enabling maximum oxygen delivery per heartbeat. (5) Evidence from pulse-rate experiments, breathing-rate observations, and capillary-refill tests all point to the same underlying mechanism: the mammalian circulatory system is tuned to rapidly match oxygen supply to metabolic demand — which is why Kipchoge's heart rate drops back to 37 bpm so quickly after a race.	By the end of this lesson, learners can explain, using a CER framework and a cumulative visual model, that: the structure of an animal's transport system directly determines its ability to meet metabolic demands and therefore survive and thrive in its environment. A grasshopper's open circulatory system is efficient for a small, low-metabolic-demand organism — haemolymph does not carry oxygen (which is delivered directly to cells via tracheae), so no high-pressure pump is needed. If Eliud Kipchoge had the circulatory system of a grasshopper, he could not deliver enough oxygenated blood to his leg muscles at marathon pace — his muscles would fail within seconds of any intense effort. His four-chambered heart, double circulation, and haemoglobin-rich blood constitute a system capable of delivering 20+ litres of oxygenated blood per minute under race conditions, then rapidly clearing metabolic byproducts and restoring resting conditions — a feat no open circulatory system could achieve.

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